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Redox mediators and microaeration boosting the linear alkylbenzene sulfonate removal

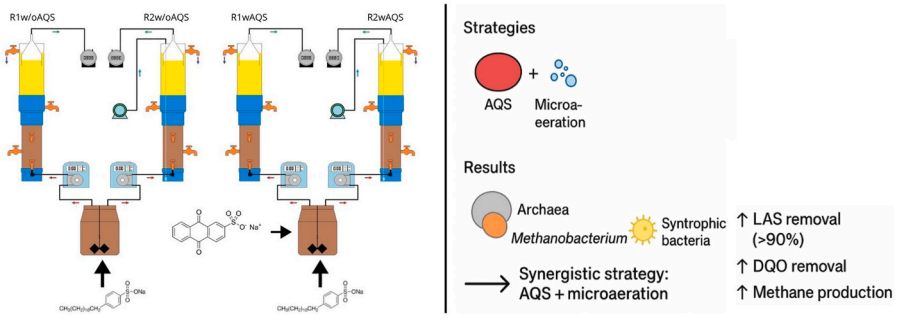
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HIGHLIGHTS

- AQS and microaeration acted synergistically, enabling LAS removal above 90% in UASB.
- Microaeration increased the specific methane yield per COD applied.
- AQS enriched *Methanobacterium*, supporting a quinone-mediated redox cycle mode.
- The combined strategy reshaped the microbial community, enriching syntrophic bacteria.
- Microaeration increased microbial richness and diversity.

GRAPHICAL ABSTRACT



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ABSTRACT

Linear alkylbenzene sulfonate (LAS) is one of the most widely used anionic surfactants and is frequently detected in wastewater, while its removal under anaerobic conditions remains limited and unstable. This study evaluated the independent and combined effects of the redox mediator anthraquinone-2-sulfonate (AQS) and controlled microaeration on LAS biodegradation in upflow anaerobic sludge blanket (UASB) reactors. Two reactors treating synthetic wastewater containing 15 mg L^{-1} LAS were operated for 90 days: R1, a strictly anaerobic UASB, and R2, a micro-aerated UASB. Each reactor was assessed without (Period I) and with AQS addition (50 μM, Period II). Under strictly anaerobic conditions without AQS, LAS removal was low and highly unstable (32 ± 16%), with pronounced performance fluctuations. The addition of AQS alone significantly improved LAS removal to 65 ± 4% without affecting methane production, indicating enhanced LAS transformation via electron shuttling rather than changes in methanogenic activity. Microaeration combined with AQS yielded the best overall performance, increasing LAS removal to 77 ± 1%, COD removal to 92.3 ± 3.8%, total methane production to 5.6 ± 0.7 L/day, and specific methane yield to 0.50 ± 0.11 $L_{CH_4} \cdot g_{COD_{app}}^{-1}$. Microaeration without AQS improved process stability but did not achieve LAS removal comparable to AQS. Microbial analyses showed that AQS promoted the enrichment of hydrogenotrophic methanogens, particularly *Methanobacterium*, and syntrophic taxa such as *midas_g_2229*, while microaeration increased microbial richness. The combined condition exhibited the highest archaeal enrichment and microbial richness. Results demonstrate that AQS and microaeration act through

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complementary mechanisms, and that their combination effectively overcomes limitations of anaerobic LAS removal.

1. Introduction

Linear alkylbenzene sulfonate (LAS) is the most widely used anionic surfactant (Palmer and Hatley, 2018; Wang et al., 2024). It is a primary ingredient in many household and industrial products, including laundry detergents, cleaning agents, and personal care products (Ran et al., 2024; Zhu et al., 2020). Because LAS is widely used and not completely removed by conventional wastewater treatment plants (WWTPs), it is often found in final effluents and in nearby water bodies. Even low concentrations in aquatic environments raise concerns about persistence, toxicity, and bioaccumulation. Research shows toxicity at levels of at least 15 μg/L (Luo et al., 2023).

LAS is biodegradable under certain conditions, but its degradation in WWTPs is often incomplete, especially under anaerobic conditions (Ogawa and Kawase, 2021). Commercial LAS mixtures are complex, containing homologues of varying alkyl chain lengths and positional isomers, which make the compounds resistant to biological treatment, particularly in the absence of oxygen (Terreros-Mecalco et al., 2024). At moderate to high concentrations, LAS can inhibit microbial activity and affect treatment performance (Wu et al., 2020). Concentrations around 2 mg/L are already sufficient to inhibit nitrifying bacteria (Dereszewska et al., 2015), while denitrifying microorganisms are inhibited at 15 mg/L (Andrade et al., 2020).

In anaerobic systems, which are increasingly used for wastewater treatment due to their energy-efficient operation and biogas production, LAS removal is typically lower and more variable than in aerobic processes (Cui et al., 2022). Their efficiency decreases when treating wastewater containing poorly biodegradable compounds, such as LAS, which can compromise effluent quality and microbial stability (Costa et al., 2024; Teran et al., 2022). Only 20% of the surfactant is biodegraded by anaerobic digestion; the low biodegradation is likely due to limited bioavailability resulting from the formation of insoluble chemical species, as well as inhibition of anaerobic biomass activity (Terreros-Mecalco et al., 2024).

Granatto et al. (2019) investigated the degradation of LAS from domestic sewage in a strictly anaerobic expanded granular sludge bed (EGSB) reactor operated at pilot scale. The authors reported an influent LAS concentration of 7.4 ± 3.8 mg L^{-1} associated with an organic loading corresponding to 527 ± 233 mg COD L^{-1} and an HRT of 36 ± 3 h, resulting in an average LAS removal efficiency of 60 ± 30%. Under these operating conditions, the anaerobic process remained microbiologically stable, with no significant changes in the microbial community structure.

Similarly, Carosia et al. (2014) evaluated LAS biodegradation in an anaerobic fluidized-bed (FB) reactor using sand as the support material and soap powder as the LAS source. In this study, the influent LAS concentration was 14 ± 3 mg L^{-1}, with an associated organic load of 532 ± 69 mg COD L^{-1} and an HRT of 15 h, resulting in an average LAS removal of 48 ± 10%. However, the higher LAS concentration exerted inhibitory effects on different microbial functional groups, as reflected in the accumulation of volatile fatty acids (VFAs), suggesting an imbalance between acidogenic and methanogenic activities and evidencing the sensitivity of anaerobic metabolic pathways to elevated surfactant loads.

Given the relevance of LAS and the importance of anaerobic treatment, strategies that enhance its degradation without affecting microbial balance or biogas production are highly valuable. In this context, the use of redox mediators has emerged as a promising approach to overcome the limitations of biodegradation of structurally complex compounds in anaerobic environments (Zhang et al., 2020).

Redox mediators are chemical compounds that enhance extracellular electron transfer between anaerobic microorganisms and electron acceptors or donors, thereby increasing the biodegradation rates of recalcitrant pollutants (do Nascimento et al., 2021b). Quinone-based mediators such as anthraquinone-2-sulfonate (AQS) and anthraquinone-2,6-disulfonate (AQDS) have attracted considerable attention due to their reversible redox behavior, low toxicity, and chemical stability under anaerobic conditions. These compounds act as electron shuttles by accepting reducing equivalents from microbial metabolism and subsequently transferring them to target contaminants, thereby accelerating reductive transformation pathways already demonstrated for dyes, chlorinated compounds, and polycyclic aromatic hydrocarbons (He et al., 2020; Zhou et al., 2018).

In the specific case of linear alkylbenzene sulfonate, AQS may interact with distinct structural moieties of the molecule, particularly the sulfonate group and the hydrophobic alkyl chain. The reduced form of AQS can facilitate electron transfer to the aromatic ring and the sulfonate substituent, promoting reductive weakening of the carbon-sulfur bond and favoring desulfonation as an initial step in LAS degradation (Wu et al., 2020; Zhang et al., 2020). In parallel, enhanced electron availability may stimulate reductive reactions along the alkyl chain, facilitating chain scission and subsequent shortening of the hydrophobic moiety, which increases molecular polarity and bioavailability (Palmer and Hatley, 2018; Van der Zee and Cervantes, 2009). By mediating these key reductive steps, AQS can lower the kinetic and thermodynamic barriers to LAS transformation under anaerobic conditions, thereby supporting more efficient conversion of LAS into intermediates that are more amenable to further anaerobic or syntrophic degradation.

Lu et al. (2021) noticed an increase of 15% of azo dye RR X–3B removal when AQS was used during the treatment of synthetic wastewater in up-flow anaerobic reactors. He et al. (2017) conducted anaerobic batch tests with synthetic wastewater containing 10 μg L^{-1} of the antibiotic sulfamethoxazole (SMX) and found that only 10 μM of AQDS enhanced contaminant removal by approximately 50%.

LAS aeration biodegradation is energy-intensive due to the high oxygen (O_2) requirements (Liu et al., 2018), and traditional direct aeration can lead to serious foaming problems (Zhang et al., 2023). In this regard, introducing low doses of oxygen via microaeration into anaerobic systems may be advantageous. Controlled oxygen inputs stimulate facultative microbial populations, promote partial oxidation of xenobiotics, and facilitate subsequent anaerobic degradation, all while maintaining conditions favorable for biogas production (Li et al., 2024).

Microaeration has shown promising results in removing recalcitrant pollutants. Collectively, previous studies have demonstrated that introducing low levels of oxygen into anaerobic systems can substantially enhance the removal of a wide range of recalcitrant and micropollutant compounds. Siqueira et al. (2018) observed that, for BTEX compounds, microaeration rates of 0.5–2.0 mL air·min^{-1} resulted in removal efficiencies above 83%, with benzene removal increasing by about 30% compared to strictly anaerobic conditions. Extending this concept to pharmaceutical compounds, Buakaew and Ratanatamskul (2024) reported that microaeration in an AnBB-MBR treating domestic wastewater markedly enhanced antibiotic removal, achieving 78% for ciprofloxacin and 88–91% for sulfamethoxazole, substantially outperforming the non-aerated reactor.

In line with these findings, Carvalho et al. (2020) showed that microaeration greatly improved the removal of recalcitrant pollutants in anaerobic treatment systems; in a microaerated upflow anaerobic sludge blanket (UASB) reactor treating textile wastewater containing the dye DB22, complete removal of aromatic amines was achieved, and effluent toxicity was reduced 16-fold compared to the strictly anaerobic reactor,

where such compounds persisted. Similarly, do Nascimento et al. (2021b) reported that, in a UASB reactor treating micropollutants such as hormones and pharmaceuticals, an airflow of 4 mL min^{-1} increased removal to 90–94%. In contrast, anaerobic treatment alone achieved less than 30%.

To the best of our knowledge, the application of either microaeration or AQS in anaerobic systems specifically for LAS removal has not yet been explored. Therefore, the objective of this study was to evaluate the effect of AQS on LAS removal in UASB reactors operated at low HRT, typical of full-scale sewage treatment in tropical regions. Operation at low HRTs increases volumetric organic and LAS loading rates, imposing stronger kinetic and redox constraints on anaerobic biomass. Under these conditions, LAS degradation is often limited by inefficient electron transfer and the accumulation of partially transformed intermediates, leading to low and unstable removal efficiencies. These limitations are intensified by the high demand for reducing equivalents associated with surfactant biodegradation. Within this framework, AQS and controlled microaeration were investigated as complementary strategies to alleviate electron transfer bottlenecks and expand the redox flexibility of the granular sludge. By facilitating extracellular electron shuttling and enabling additional redox pathways, these approaches were hypothesized to enhance LAS transformation and improve reactor stability under realistic low-HRT conditions.

2. Materials and methods

2.1. Experimental set-up

Two lab-scale UASB reactors (working volume of 3.5 L) (Fig. 1), inoculated with anaerobic sludge [Total solids (TS) = 50 ± 3 g L^{-1}, total volatile solids (TVS) = 28 ± 1 = g·L^{-1} and total fixed solids (TFS) = 23 ± 1 g L^{-1}], were operated, in parallel, at an HRT of 8 h and room temperature of approximately 28 °C.

Both reactors operated under the same conditions except for microaeration. R1 was a conventional UASB reactor, while R2 included microaeration and was called UMSB (Upflow Microaerobic Sludge Blanket). The experiment had two 45-day periods, differing only in the addition of AQS. In Period I, no AQS was added ($R1_{AQS-}$ and $R2_{AQS-}$). In Period II, both systems ($R1_{AQS+}$, $R2_{AQS+}$) were supplemented with 50 μM AQS (98%, Sigma-Aldrich, USA) to evaluate its impact on system performance. This structured approach allowed for a systematic comparison of the effects of AQS matter degradation and LAS removal under both anaerobic and microaerobic conditions.

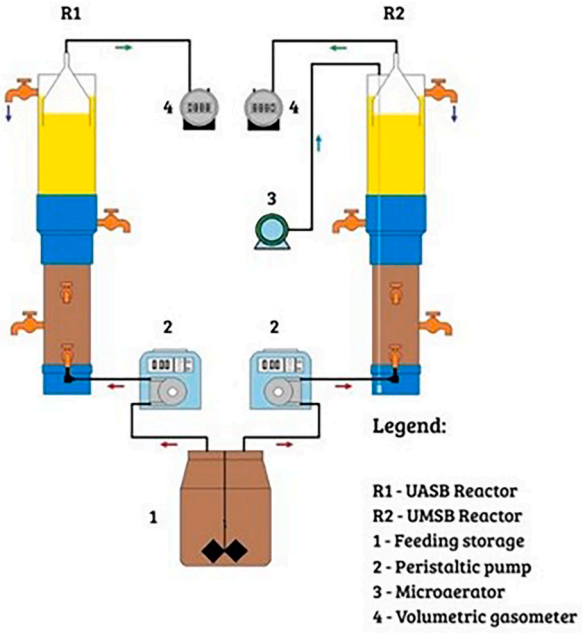


Fig. 1. Schematic of the continuous experimental set-up.

Microaeration in R2 was performed with synthetic air (80% N_2:20% O_2, White Martins, Brazil) at its base by means of a mass flow controller (GFC17, Cole-Parmer, USA). The microaeration air dose used in R2 was approximately 1.0 L_{air}·min^{-1}, corresponding to 0.2 LO_2·L_{feed}^{-1}·d^{-1}, which was kept constant during the experimental period.

The microaeration rate and the AQS concentration were defined based on previous studies conducted by our research group, in which these values proved effective in stimulating electron transfer processes in aromatic compounds, without causing oxidative stress or toxicity to the anaerobic biomass (de Barros et al., 2018; Buarque et al., 2019).

The reactors were fed with synthetic wastewater containing the LAS (Sodium dodecylbenzenesulfonate, 100%, Sigma-Aldrich, USA) (15 mg L^{-1}), sucrose (1 gCOD·L^{-1}), basal medium (macro and micronutrients) prepared according to Firmino et al. (2010), and sodium bicarbonate (1 g L^{-1}) to maintain the pH close to 7.0. All reagents were used as purchased without further purification. The synthetic influent was kept refrigerated at 4 °C to prevent natural biodegradation of organic matter.

2.2. Chemical and chromatographic analysis

Chemical oxygen demand (COD), pH, TSS, and TS content were determined according to APHA (2012).

The quantification of CH_4, CO_2, H_2, and H_2S in the biogas was determined by gas chromatography with ionization detection by dielectric barrier discharge (GC BID-2010 Plus, Shimadzu Corporation, Japan), equipped with a GS-GASPRO column (60 m × 0.32 mm) (Agilent Technologies Inc., USA). Helium gas was used as the carrier gas (White Martins LTDA, Brazil) at a flow rate of 2 mL min^{-1}, with a run time of 9 min. Oven, injector, and detector temperatures were 50, 100, and 250 °C, respectively. O_2 and N_2 were quantified by gas chromatography with thermal conductivity detection (GC-TCD) (GC-17A, Shimadzu Corporation, Japan). The biogas sample (1.0 mL) was injected in splitless mode, and chromatographic separation was performed on a Mol Sieve 5A PLOT column (30 m, 0.32 mm ID) (Restek Corporation, USA). Oven, injector, and detector temperatures were 35, 40, and 230 °C, respectively. Helium (White Martins, Brazil) was used as the carrier gas at a flow rate of 7 mL min^{-1}, and the run time was 5 min.

LAS concentrations were determined by high-performance liquid chromatography (HPLC) using a Shimadzu system (CBM-20A, SPD-20A, CTO-20A, SIL-20A, and LC-20AT) equipped with a reversed-phase C18 column from Shimadzu, as described by Silva et al. (2017). The mobile phase was water with 5 mmol L^{-1} ammonium acetate (solvent A) and acetonitrile (solvent B); the mobile phase flow rate was 1 mL min^{-1}, the sample volume was 50 μL, the temperature of the thermostated compartment of the column was 40 °C, and UV-Vis detection at 230 nm. A gradient was applied to the mobile phase to separate the LAS constituents; it started at 45% solvent B, which began to rise after 5 min and reached 60% in 20 min, then returned to 45% after 1 min and remained at 45% for 6 min.

Biomass samples from the inoculum and the four reactors were sent to identify microbial communities and underwent extraction, sequencing, and analysis of absolute abundances. DNA collection, extraction, 16S rRNA gene amplicon sequencing, and data processing were performed following the methodology described by Silva et al. (2023). The sequencing data generated in this study have been submitted to the National Center for Biotechnology Information (NCBI) BioProject repository under the ID PRJNA836880.

2.3. Statistical analysis

The Mann-Whitney and Kruskal-Wallis nonparametric tests, which do not require specific distributional assumptions, were used to compare reactor performance across the different experimental periods at the 5%

significance level.

3. Results and discussion

3.1. Performance of the UASB reactors

3.1.1. Operational parameters and volumetric biogas production

The evaluation of operational pH and acidity parameters in the UASB reactors, both with and without microaeration, revealed very similar values, regardless of AQS use (Fig. 2). The average pH remained between 7.4 and 7.6 throughout the experimental period, with no significant differences between reactors. Acidity/TA ratios ranged from 0.54 ± 0.09 to 0.82 ± 1.06, and during the stabilization phase, they were further stabilized, with all values falling below the threshold of 0.8 typically associated with process imbalance.

Therefore, the application of AQS and controlled microaeration did not cause acid-base disturbances or instability in either pH or Acidity/TA. This suggests that these strategies are compatible with the physicochemical equilibrium required for anaerobic treatment systems and do not compromise reactor stability when applied at moderate doses. In the study by do Nascimento et al. (2021b), microaeration (1 mL air·min^{-1}) combined with the redox mediator AQDS (50–100 μM) also maintained system stability, with the oxidation-reduction potential (ORP) remaining below −200 mV, ensuring conditions suitable for reductive biotransformation.

Fig. 3 presents the volumetric production of biogas and methane for each UASB reactor over the experimental period. Reactors $R1_{AQS-}$ and $R1_{AQS+}$ (anaerobic conditions, without and with AQS) showed similar patterns ($p = 0.064$), with average methane yields of 2.0 ± 0.2 L/day and 1.9 ± 0.3 L/day, respectively. The methane fraction in the biogas from these reactors was stable and consistent with expected values for anaerobic digestion of readily biodegradable substrates, averaging 67.2 ± 2.7% ($R1_{AQS-}$) and 67.6 ± 2.7% ($R1_{AQS+}$) ($p = 0.026$). These results indicate a predominance of methanogenic activity and suggest that adding AQS under strictly anaerobic conditions did not adversely affect methane production.

In contrast, reactors $R2_{AQS-}$ and $R2_{AQS+}$, which received microaeration (without and with AQS), produced significantly higher volumes of biogas and methane ($p < 0.05$). The total volumetric methane production reached 4.9 ± 0.6 L/day in R2 and 5.6 ± 0.7 L/day in $R2_{AQS+}$, indicating a substantial increase in overall gas production compared to anaerobic treatments. However, these increases were accompanied by a marked reduction in the methane content of the biogas, which averaged only 31.9 ± 4.3% ($R2_{AQS-}$) and 33.5 ± 4.3% ($R2_{AQS+}$). These values are less than half of those observed in the strictly anaerobic reactors.

This inverse relationship between total biogas production and methane concentration is more consistently explained by the presence of oxygen in the microaerated reactors, which altered gas composition through multiple coupled mechanisms rather than solely by nitrogen dilution. The supplied oxygen was likely consumed by facultative and aerobic microorganisms, promoting partial oxidation of reduced compounds and increasing CO_2 formation, thereby reducing the volumetric fraction of methane in the biogas. In addition, oxygen availability may have redirected part of the electron flow away from strict methanogenesis toward alternative respiratory pathways, thereby affecting methane yield without necessarily reducing overall biogas generation.

In this context, Buarque et al. (2019) highlighted that low oxygen concentrations in anaerobic systems promote partial oxidation of substrates, favoring biomass growth and alternative metabolic pathways while decreasing methane production efficiency. Consistent with this trade-off between pollutant removal and energy recovery, several studies have reported that microaeration can enhance contaminant degradation at the expense of methane yield. For example, Guimarães de Oliveira et al. (2021), evaluating the effect of microaeration in a UASB reactor treating swine wastewater (2.0 mL air·L feed^{-1}), observed that no methane was detected in the biogas from the UMSB reactor, a result attributed to methane dilution by other gases due to the continuous injection of atmospheric air. Despite the reduction in methane content, the increase in absolute methane volume in $R2_{AQS-}$ and $R2_{AQS+}$ suggests that methanogenesis was not suppressed. Instead, a portion of the organic load was likely directed to oxidative metabolism or assimilated into new biomass, while the remainder continued to be converted into methane.

The application of AQS in $R2_{AQS+}$, when combined with microaeration, resulted in similar methane yields across all treatments ($p > 0.05$). This behavior is consistent with previous findings by do Nascimento et al. (2021b), who demonstrated that the introduction of microaeration in UASB reactors increases total biogas production while reducing the methane volumetric fraction due to gas dilution, without compromising the biogas energetic quality. Their results support the interpretation that microaeration modifies biogas composition primarily through changes in gas balance rather than through inhibition of methanogenic activity, consistent with the trends observed in the present study.

Moreover, the decrease in methane concentration does not necessarily indicate process instability or inhibition of anaerobic metabolism. The observed shift in biogas composition is better interpreted as a combined effect of methane dilution by increased CO_2 production and the potential for biological methane oxidation in microaerobic niches within the reactor. These processes are consistent with the gas composition data and suggest that microaeration primarily modified carbon flow and redox balance rather than compromising the stability of the anaerobic treatment process.

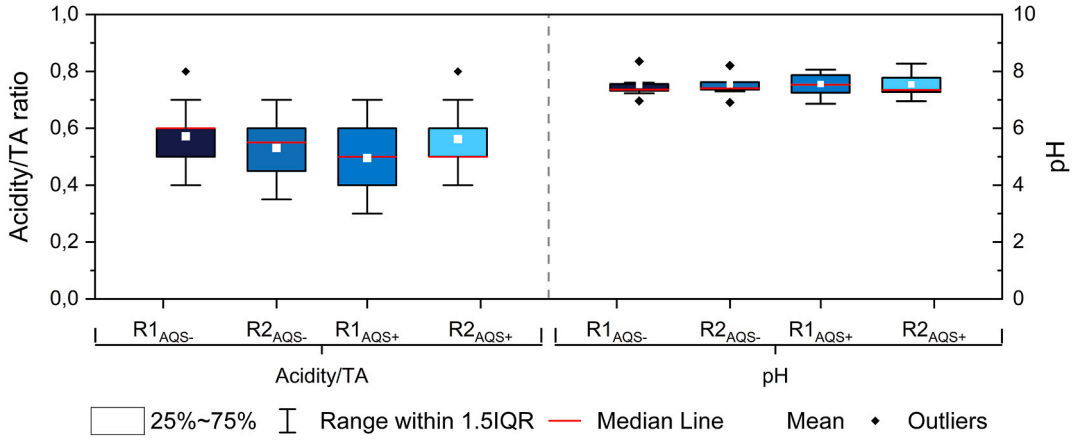


Fig. 2. pH and Acidity/TA ratio during acclimation and stabilization phases for UASB without AQS ($R1_{AQS-}$), microaerated UASB without AQS ($R2_{AQS-}$), UASB with AQS ($R1_{AQS+}$) and microaerated UASB with AQS ($R2_{AQS+}$).

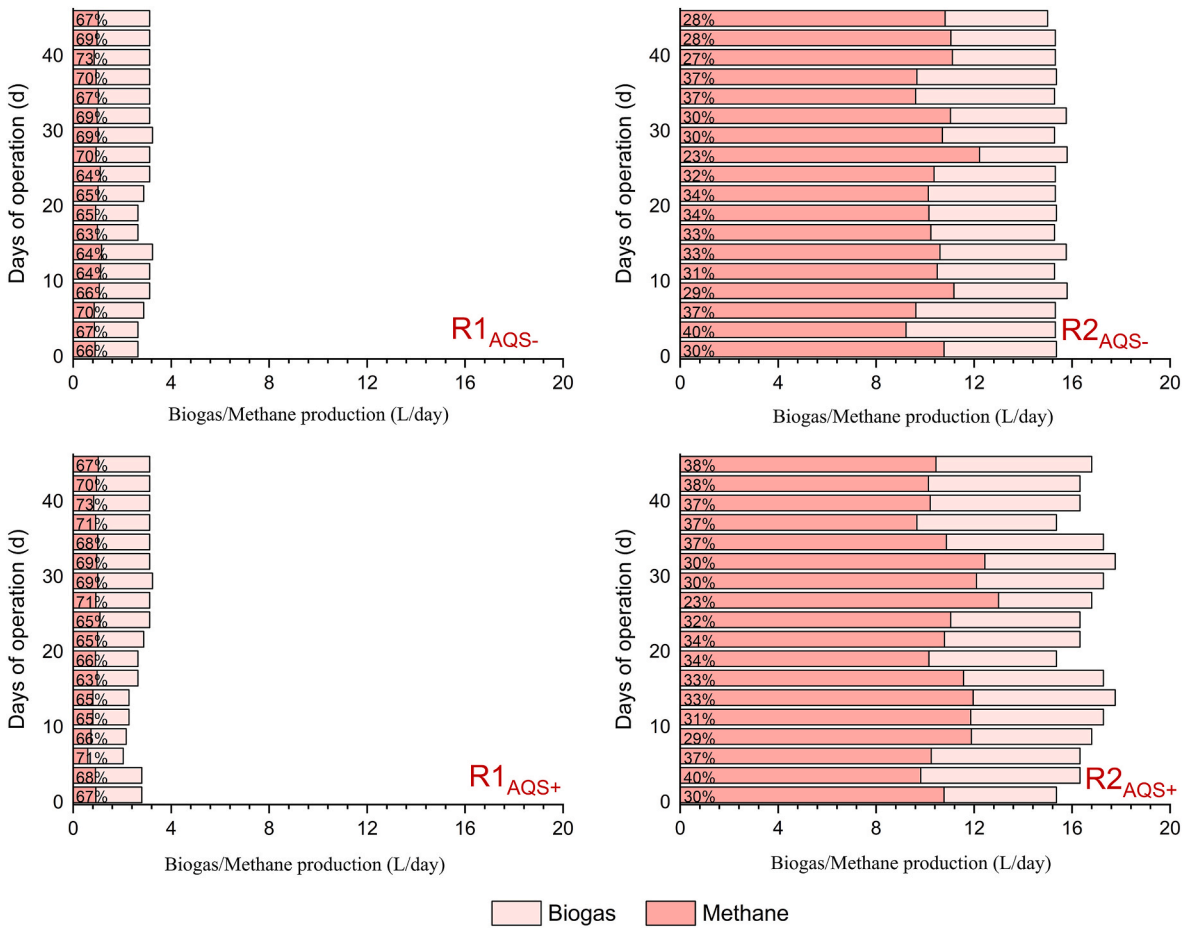


Fig. 3. Biogas and methane production from reactors throughout the periods without AQS ($R1_{AQS-}$ and $R2_{AQS-}$) and with AQS ($R1_{AQS+}$ and $R2_{AQS+}$). The percentages indicated in the bars refer to the percentage of methane in the biogas.

At the same time, the presence of a redox mediator does not affect the production and/or composition of biogas. In this context, while volumetric methane production can be maintained or even increased, the biogas quality, in terms of methane concentration, may be compromised. Further research into optimizing oxygen dosing and mediator concentration could help balance these outcomes, aiming for enhanced pollutant removal without significant loss in methane purity.

3.1.2. *Efficiency parameters*

The COD removal efficiency and the specific methane yield per unit of applied organic load are presented in Fig. 4. All reactors achieved high COD removal efficiencies, exceeding 80%, indicating effective conversion of organic matter under all operational conditions. In reactors $R1_{AQS-}$ and $R1_{AQS+}$ (anaerobic, with and without AQS), average COD removals were 82.3 ± 8.0% and 84.9 ± 7.3%, respectively, showing that the addition of AQS alone did not significantly impact the degradation of bulk organic matter ($p > 0.05$).

Reactors $R2_{AQS-}$ and $R2_{AQS+}$, which received microaeration, presented higher COD removal efficiencies ($p < 0.05$) of 86.5 ± 4.3% and 92.3 ± 3.8%, respectively. These results suggest that introducing low

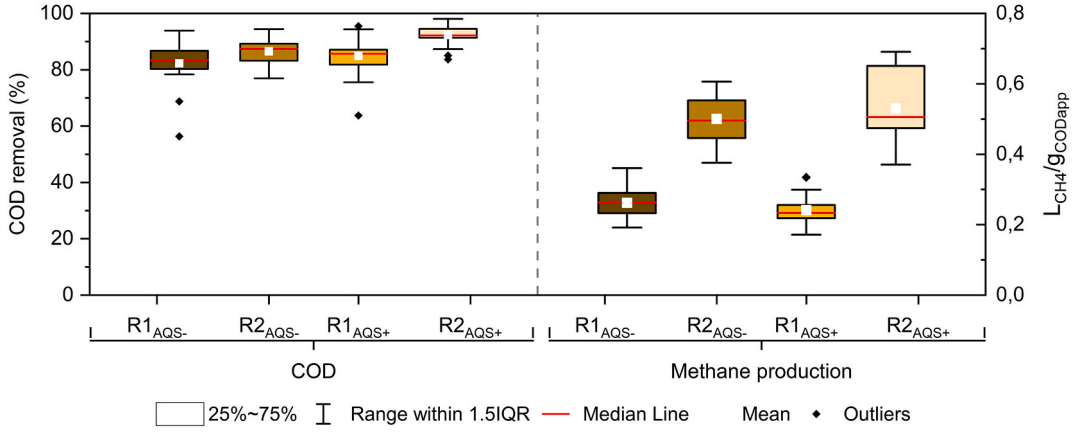


Fig. 4. pH and Acidity/TA ratio in anaerobic (R1) and microaerobic (R2) systems with ($R1_{AQS+}$, $R2_{AQS+}$) and without ($R1_{AQS-}$, $R2_{AQS-}$) the redox mediator AQS.

oxygen doses enhanced the system's oxidative degradation capacity, particularly when combined with AQS in $R2_{AQS+}$. The elevated performance in $R2_{AQS+}$ may result from synergistic effects between oxygen availability and redox mediation, which collectively stimulate both aerobic and anaerobic metabolic routes, favoring the degradation of more complex or partially oxidized intermediates (Buakaew and Ratanatamskul, 2024; do Nascimento et al., 2021b).

However, when COD removal is analyzed alongside methane production, a distinct trend emerges. The specific methane production ($L_{CH4}/gCOD_{app}$) was significantly lower ($p < 0.05$) in anaerobic reactors (0.19 ± 0.05 in $R1_{AQS-}$ and 0.17 ± 0.04 in $R1_{AQS+}$) compared to the microaerated systems (0.46 ± 0.08 in $R2_{AQS-}$ and 0.50 ± 0.11 in $R2_{AQS+}$).

This apparent contradiction can be explained by the fact that while oxygen presence reduced the methane percentage in the biogas, it did not reduce total methane generation relative to organic load. On the contrary, microaeration likely stimulated faster and more complete substrate degradation, leading to greater methane volumes per gram of COD applied, especially in $R2_{AQS+}$, where the AQS may have facilitated redox reactions essential for the breakdown of intermediate products. Similar effects have been reported in anaerobic reactors treating antibiotics and aromatic compounds using AQS and AQDS, where the redox mediator enhanced electron transfer and accelerated rate-limiting steps (Buakaew and Ratanatamskul, 2024; dos Santos et al., 2007).

Furthermore, these results suggest that although a portion of the COD may have been consumed via oxidative pathways or biomass synthesis under microaerobic conditions, a substantial fraction still followed the methanogenic route, particularly in the presence of AQS. The performance of $R2_{AQS+}$ thus highlights the potential of combining microaeration and redox mediation to achieve both high removal efficiency and energy recovery in UASB systems.

Finally, the greater consistency observed in the boxplots for both COD removal and methane yield in $R2_{AQS+}$ supports the reproducibility of this combined strategy. Compared to $R1_{AQS-}$ and $R1_{AQS+}$, the lower variability in $R2_{AQS+}$ may indicate a more stable and resilient microbial community, better adapted to simultaneous oxygen exposure and redox cycling – an aspect further explored in Section 3.3.

3.2. Removal of linear alkylbenzene sulfonate

3.2.1. Effect of microaeration

In the first days of operation, the four reactors showed average LAS removal efficiencies of around $79 \pm 1\%$, with no statistically significant differences between the systems ($p > 0.05$) (Fig. 5). This initial behavior can be attributed to the greater functional diversity of the newly established anaerobic microbiota, as populations are not yet under the selective pressure of sulfonated compounds. At this stage, the lower accumulation of toxic intermediates and the absence of prolonged stress are likely to favor greater activity of consortia capable of degrading xenobiotic compounds such as LAS, even if only partially.

In addition, LAS concentrations had not yet reached critical inhibition levels, which may have allowed secondary or nonspecific metabolic pathways to act in its initial transformation. Among these mechanisms, the action of nonspecific oxidoreductases stands out, which, according to Zhou et al. (2022) and Costa et al. (2024), can actively participate in the biodegradation of surfactants such as LAS by promoting oxidation and reduction reactions that are not very selective but effective in initially breaking the aromatic structure and aliphatic chain of these compounds. Similarly, Li et al. (2024) observed that UASB reactors showed high efficiency in removing sodium dodecylbenzene sulfonate (a type of LAS) during the first days of operation, with initial values around 80%, which decreased to less than 20% after 120 days.

Throughout the operation, reactors not subjected to microaeration ($R1_{AQS-}$ and $R1_{AQS+}$) exhibited greater operational instability in LAS removal, with more pronounced fluctuations in degradation efficiency and a progressive downward trend, especially in the system without AQS ($R1_{AQS-}$). $R1_{AQS-}$ showed the worst performance in LAS removal, with a continuous downward trend until reaching final valuesclose to 11%. This instability may be related to the metabolic limitation imposed by a strictly anaerobic environment in the face of highly recalcitrant compounds, such as LAS. In the absence of oxygen, LAS degradation depends exclusively on less efficient anaerobic pathways, which are often incapable of breaking the sulfonated aromatic ring, leading to the accumulation of toxic intermediates or the interruption of metabolic pathways (Corada-Fernández et al., 2018). The lack of microaeration also compromises the development of niches for facultative or microaerophilic microorganisms (Castro et al., 2023; do Nascimento et al., 2021a), which could complement surfactant degradation through partial oxidative processes, thereby contributing to greater system stability.

The systems subjected to microaeration showed significantly higher and more stable LAS removals over time than reactors operated under strictly anaerobic conditions ($p < 0.001$). In particular, the $R2_{AQS-}$ reactor, microaerated and without a redox mediator, achieved an average removal of $70 \pm 3\%$, while the $R1_{AQS-}$ reactor, operated without microaeration and without AQS, showed lower removal, with an average of $32 \pm 16\%$. These results confirm that the presence of oxygen, even at very low concentrations, is decisive for the degradation of surfactants such as LAS, corroborating the literature that points to the aerobic pathway as the main transformation route of this compound (Costa et al., 2024).

The controlled introduction of oxygen into the system favors the activity of strict and facultative aerobic microorganisms, which can express oxygenases, especially monooxygenases, that catalyze the initial hydroxylation of the aromatic ring of LAS (Fu et al., 2023; Xu et al.,

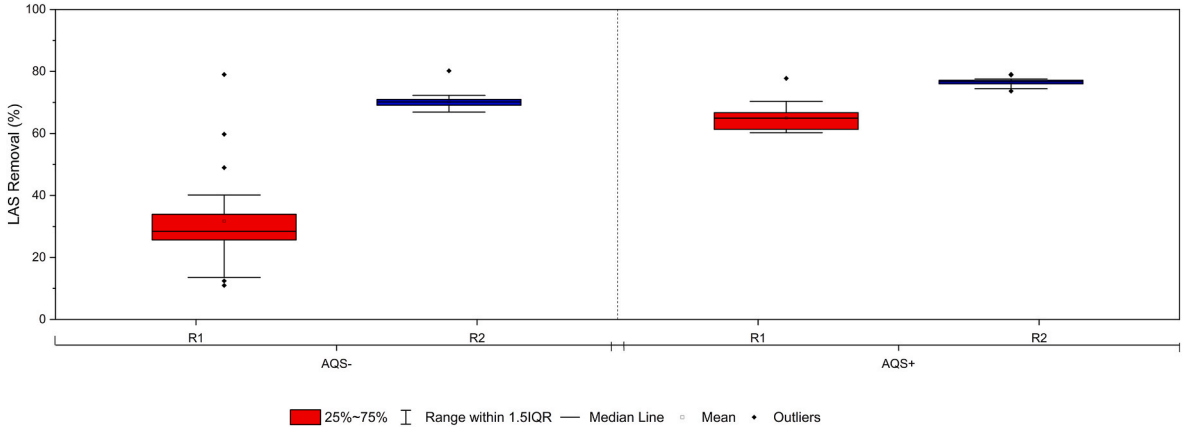


Fig. 5. LAS removal in anaerobic (R1) and microaerobic (R2) systems with ($R1_{AQS+}$, $R2_{AQS+}$) and without ($R1_{AQS-}$, $R2_{AQS-}$) the redox mediator AQS.

2023). This reaction transforms the original compound into more hydrophilic and reactive derivatives, paving the way for the cleavage of the aromatic ring and, subsequently, for the complete or partial mineralization of the molecule. According to Zhou et al. (2022), monooxygenation requires molecular oxygen as a terminal electron acceptor, which explains the significant improvement in LAS removal under microaeration.

Furthermore, the partial oxidation promoted under these conditions yields less toxic, more biodegradable intermediates, thereby reducing the inhibitory impact of LAS on biomass.

The introduction of microaeration into anaerobic systems has been reported to enhance the biodegradation of surfactants and other difficult-to-biodegrade compounds. Cheng et al. (2018) reduced the surfactant concentration from 9000 mg/L to 2000 mg/L without foam formation or significant VFAs mineralization. Carvalho et al. (2020) achieved complete removal of the azo dye Direct Black 22 with microaeration in the UASB reactor.

The action of microaerobic pathways also contributes to maintaining a more dynamic redox environment, in which fermentative, reducing, and oxidative processes coexist, increasing metabolic diversity and, consequently, the system's resilience in the presence of recalcitrant compounds (do Nascimento et al., 2021b; Fu et al., 2023). Therefore, microaeration promotes not only greater efficiency but also greater stability in LAS removal, as evidenced by the low variability in results obtained in $R2_{AQS-}$.

3.2.2. Impact of anthraquinone-2-sulfonate (AQS) supplementation

The addition of the redox mediator anthraquinone-2-sulfonate (AQS) resulted in significant increases in LAS removal in both systems, demonstrating the fundamental role of this compound in enhancing electron transfer and overcoming limiting steps in LAS biodegradation. Despite this, the microaerated reactor with AQS ($R2_{AQS+}$) showed the best overall performance, with an average removal of 77 ± 1%, significantly higher than that observed in the reactor also doped with AQS, but operated without microaeration ($R1_{AQS+}$), which obtained 65 ± 4% ($p < 0.001$).

AQS functions as an extracellular redox mediator, promoting electron transfer between microorganisms and substrates and yielding very satisfactory results in systems treating highly recalcitrant compounds, such as LAS (Ren et al., 2022; Song et al., 2020). The beneficial effect of quinone-based redox mediators is not restricted to inorganic or surfactant pollutants. For instance, do Nascimento et al. (2021b) observed an increase in antibiotic removal from 6% to 75% upon the addition of such mediators. In line with these findings, Tang et al. (2024) also reported an improvement in the denitrification rate in the presence of quinone-type redox mediators.

In this context, AQS is highly stable and can alternate between its oxidized (quinone) and reduced (hydroquinone) forms, facilitating redox reactions in conditions where microorganisms have limited direct electron transfer (Ren et al., 2022). Therefore, during LAS degradation, the presence of AQS can facilitate desulfonation and promote aromatic ring breakdown, favoring the formation of metabolites more accessible to anaerobic and facultative aerobic pathways.

This behavior becomes evident when the operational performance of the different reactors is examined comparatively. Under strictly anaerobic conditions, AQS addition increased LAS removal without substantially affecting methane production, indicating that the redox mediator primarily facilitated contaminant transformation rather than altering the overall digestion pathway. Conversely, microaeration alone improved reactor stability and organic matter removal but led to fluctuating LAS removal at lower levels. When both strategies were applied simultaneously in reactor $R2_{AQS+}$, higher LAS removal was achieved, along with increased methane production and improved COD removal. This concurrent improvement in multiple performance indicators suggests that microaeration and AQS influenced complementary stages of the LAS degradation process.

Low oxygen availability likely promoted initial activation or conditioning of the surfactant structure, while AQS enhanced subsequent reductive steps by facilitating electron transfer, thereby preventing the accumulation of inhibitory intermediates. As a result, the combined operation led to a process response that exceeds what would be expected from the individual effects of microaeration or AQS alone, supporting the interpretation of a synergistic interaction at the process level.

Although AQS promoted significant gains in LAS removal in both systems, its effect was more pronounced in the conventional UASB reactor operated without microaeration. In $R1_{AQS+}$, the presence of the mediator resulted in a removal efficiency about twice that of the control reactor without AQS ($R1_{AQS-}$), increasing the average from 32 ± 16% to 65 ± 4%. On the other hand, in the system with microaeration ($R2_{AQS+}$), the increase provided by the AQS was proportionally smaller, reaching only about 10% compared to the microaerated reactor without mediator ($R2_{AQS-}$), which achieved 70 ± 3%. These results indicate that although the AQS significantly intensified LAS biodegradation, especially under strictly anaerobic conditions, it was not sufficient to match the removal levels achieved by systems with oxygen supply ($p = 0.017$).

The difference observed between the systems reinforces the limitation imposed by the absence of oxygen in strictly anaerobic environments, which restricts the action of oxidative pathways essential for LAS degradation. Although the AQS increased electron transfer and promoted partial transformation of the molecule, the opening of the aromatic ring and desulfonation, which are key steps in its mineralization, remain strongly dependent on reactions that require molecular oxygen. Thus, even with the addition of the mediator, the performance of the $R1_{AQS+}$ reactor did not match that of the microaerated system without AQS ($R2_{AQS-}$), whose greater LAS removal highlights the central role of microaeration in enabling more effective biochemical pathways for recalcitrant compounds such as sulfonated surfactants.

do Nascimento et al. (2021b) also observed significant gains in antibiotic removal when combining AQS with microaeration in UASB reactors. However, the authors point out that, from a technical standpoint, diluting biogas and the associated costs may not justify its use.

In summary, the absence of microaeration prevented the formation of intermediate redox zones, thereby hindering the partial oxidation or desulfonation of LAS. Even with the presence of AQS, which favored mediated electron-transfer pathways and were essential for accelerating the rate-limiting steps of biodegradation, the absence of microaeration prevented the formation of intermediate redox zones. Therefore, although AQS represents an important technological resource for increasing the efficiency of anaerobic reactors, its effect is limited when isolated from a more favorable redox environment. The persistence of LAS in anaerobic systems, even in the presence of redox mediators, underscores the need for integrated strategies, such as combining microaeration with AQS addition, to efficiently treat recalcitrant anionic surfactants in UASB reactors.

3.3. Microbial community composition

3.3.1. Microbial community richness and diversity

Microbial richness and diversity were evaluated using Chao1 and ACE as estimators of species richness and Shannon and inverse Simpson indices as complementary measures of diversity, integrating richness and evenness. As shown in Fig. 6, the observed patterns indicate that microaeration and AQS affected microbial communities differently, helping explain the contrasting behaviors observed in LAS removal.

The markedly higher Chao1 and ACE values in $R2_{AQS-}$ and $R2_{AQS+}$ compared to the strictly anaerobic reactors indicate that microaeration was the dominant driver of species recruitment. The introduction of oxygen, even at low levels, likely promoted niche diversification by allowing the coexistence of strict anaerobes, facultative, and microaerophilic populations (Buarque et al., 2019; do Nascimento et al., 2021b). This expansion of ecological niches increases the probability of functional redundancy and metabolic complementarity, particularly

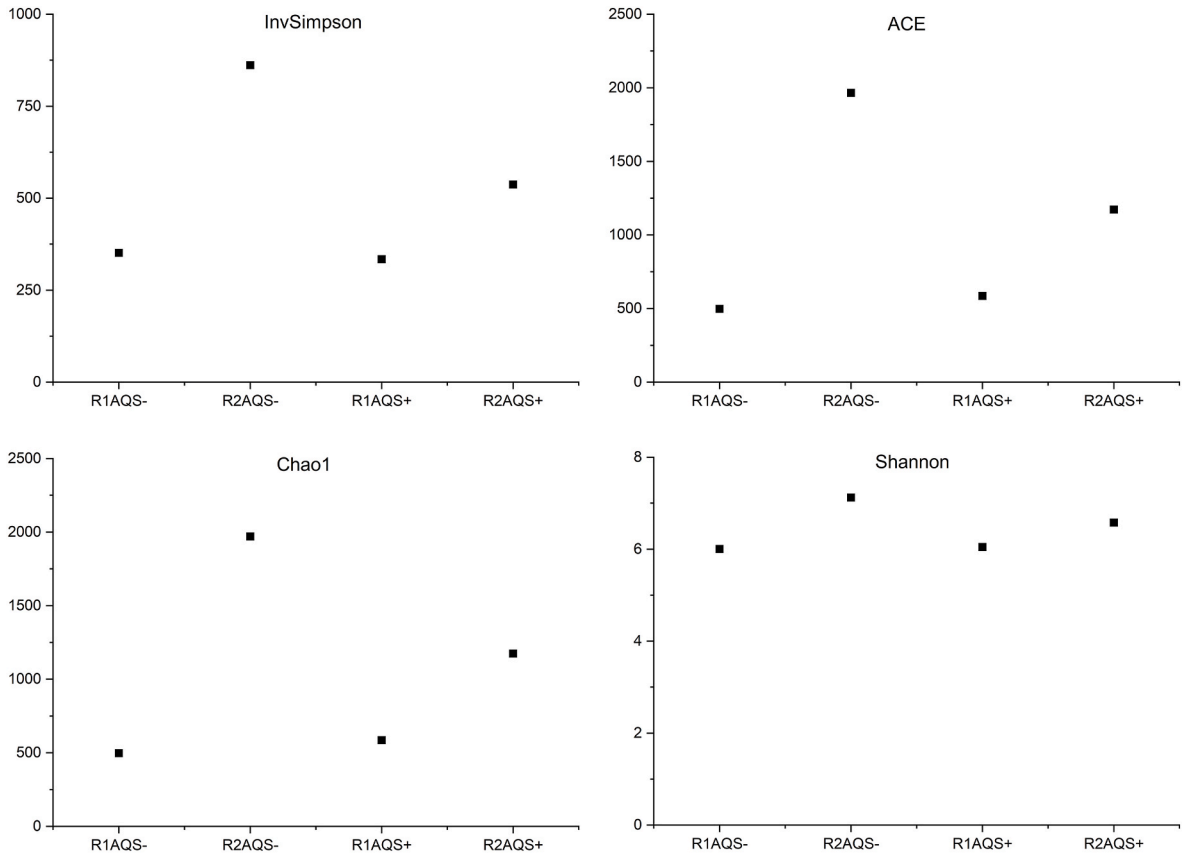


Fig. 6. Ecological indices of richness and diversity in UASB without AQS ($R1_{AQS-}$), microaerated UASB without AQS ($R2_{AQS-}$), UASB with AQS ($R1_{AQS+}$) and microaerated UASB with AQS ($R2_{AQS+}$).

relevant for the degradation of structurally complex compounds such as LAS, which require sequential and syntrophic transformations.

The consistently lower richness observed in $R2_{AQS+}$ relative to $R2_{AQS-}$ suggests that the presence of AQS imposed an additional selective pressure, favoring taxa capable of efficiently exploiting mediated electron transfer rather than promoting broad diversification.

Although Shannon indices were similar between $R2_{AQS-}$ and $R2_{AQS+}$, the inverse Simpson index clearly differentiated these reactors, indicating distinct community organization patterns. The similarity in Shannon values suggests that overall diversity, when weighted toward rare taxa, remained comparable. In contrast, the lower inverse Simpson value in $R2_{AQS+}$ indicates reduced evenness and increased dominance of specific populations. This pattern is consistent with the enrichment of functionally specialized groups, likely including methanogenic archaea and bacteria involved in redox-active metabolisms, which become competitively favored in the presence of an efficient electron shuttle such as AQS. Therefore, while richness was preserved, community structure shifted toward dominance by taxa with higher energetic efficiency under mediated electron transfer conditions.

Under strictly anaerobic conditions, both Shannon and inverse Simpson indices were lower and largely unaffected by AQS addition, reinforcing that, in the absence of microaeration, AQS primarily acted at the metabolic level rather than reshaping community diversity. In these reactors, LAS degradation was likely constrained by kinetic and thermodynamic limitations, and the redox mediator alone was insufficient to trigger broad ecological restructuring.

The combined analysis of diversity indices suggests that enhanced LAS removal in $R2_{AQS+}$ cannot be attributed solely to increased richness or generalized microbial adaptation. Instead, the data support a scenario in which AQS selectively amplified the activity and relative abundance of key functional groups that enable coupling of LAS transformation to efficient electron transfer pathways. By facilitating electron flow from LAS-derived intermediates toward terminal electron acceptors, AQS likely reduced energetic bottlenecks associated with sulfonate reduction and alkyl chain transformation, conferring a competitive advantage to specialized degraders. Consequently, LAS removal was improved through a synergy between microaeration-driven niche expansion and AQS-mediated functional selection, rather than through a simple increase in overall microbial diversity, as observed in other studies (do Nascimento et al., 2021b; Li et al., 2022).

3.3.2. *Microbial community structure*

The microbial community structure exhibited consistent, process-relevant shifts across the reactors, closely mirroring the observed differences in LAS removal, methane production, and overall process stability (Fig. 7 and 8). At the domain level, Bacteria predominated in all systems, reflecting their central role in hydrolysis, fermentation, and the initial transformation of complex substrates such as LAS. However, pronounced variations in the relative abundance of Archaea highlight that AQS addition and microaeration exerted strong selective pressures on electron flow and syntrophic organization, rather than merely increasing microbial diversity.

In the strictly anaerobic reactor without AQS ($R1_{AQS-}$), Bacteria accounted for 72.0% of the community. Despite presenting the highest methane fraction in the biogas, this reactor showed the poorest and most unstable LAS removal. At the phylum level, *Firmicutes* and *Actinobacteriota* dominated, while at the family and genus levels, *Bogoriellaceae* and *Georgenia* were prominent. These taxa are typically associated with hydrolytic and fermentative metabolism and, in some cases, facultative behavior. Their dominance suggests a community well adapted to conventional anaerobic digestion but poorly structured to sustain reductive LAS transformation, which explains the strong fluctuations and limited

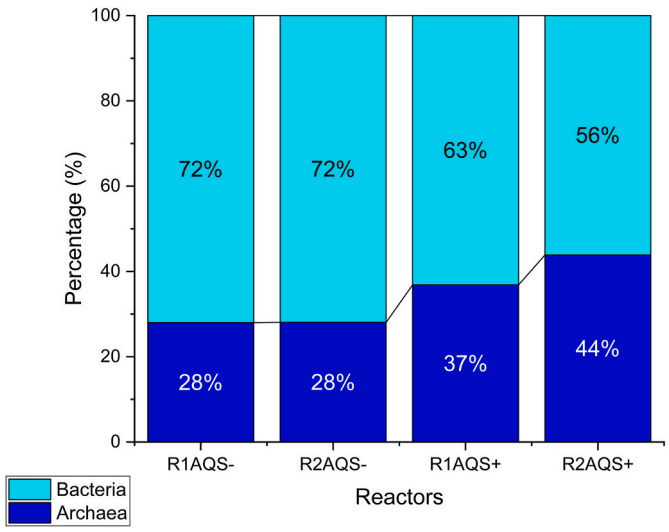


Fig. 7. Taxonomic distribution of DNA sequence in UASB without AQS ($R1_{AQS-}$), microaerated UASB without AQS ($R2_{AQS-}$), UASB with AQS ($R1_{AQS+}$) and microaerated UASB with AQS ($R2_{AQS+}$).

LAS removal observed under strict anaerobiosis.

The addition of AQS under strictly anaerobic conditions ($R1_{AQS+}$) led to a marked enrichment of Archaea, increasing their relative abundance to 36.9%, while methane production remained unchanged. At the phylum and class levels, *Euryarchaeota* and *Methanobacteria* increased substantially, indicating stimulation of hydrogenotrophic methanogenesis. Concomitantly, *Firmicutes*, particularly members of *Clostridia*, and *Actinobacteriota* remained abundant. This taxonomic configuration indicates strengthened syntrophic coupling between fermentative bacteria and methanogens, facilitated by AQS-mediated electron shuttling. Under these conditions, LAS removal increased from approximately 32% to 65%, suggesting that AQS primarily improved the thermodynamic feasibility of key reductive steps, such as desulfonation and alkyl chain transformation, without requiring an increase in overall methanogenic flux (do Nascimento et al., 2021b; Van der Zee and Cervantes, 2009).

Microaeration alone ($R2_{AQS-}$) increased richness and promoted the coexistence of strict anaerobes and facultative microorganisms, consistent with previous reports (Buarque et al., 2019; do Nascimento et al., 2021b). At the phylum level, *Euryarchaeota* became more prominent, while at the family level, *Anaerolineaceae* and *Eubacteriaceae* increased in relative abundance, reflecting reinforcement of syntrophic and fermentative networks. However, despite improved COD removal and increased total biogas production, LAS removal remained unstable. This indicates that microaeration enhanced metabolic flexibility and resilience but, in the absence of a redox mediator, did not sufficiently direct electron flow toward LAS-specific reductive pathways.

The synergistic effect of AQS and microaeration was most evident in R2AQS+, which exhibited the highest archaeal abundance (43.9%) and the best overall process performance. At the phylum and class levels, *Euryarchaeota* and *Methanobacteria* dominated, with *Methanobacteria* reaching approximately 40.0%. At the family and genus levels, *Methanobacteriaceae* and *Methanobacterium* were strongly enriched, highlighting the central role of hydrogenotrophic methanogenesis. The rapid consumption of hydrogen by these archaea is critical for maintaining low redox potential and preventing feedback inhibition of fermentative and syntrophic bacteria, thereby sustaining continuous LAS transformation (Li et al., 2022).

In parallel, the enrichment of syntrophic bacterial families such as *Anaerolineaceae* and *Eubacteriaceae*, along with the dominance of the unclassified genus *midas g_2229* in $R1_{AQS+}$ and $R2_{AQS+}$, indicates the establishment of a highly integrated syntrophic network. These taxa are frequently associated with the anaerobic degradation of long-chain hydrocarbons and surfactants and rely on close metabolic coupling with hydrogenotrophic methanogens (do Nascimento et al., 2021b; Van der Zee and Cervantes, 2009; Motteran et al., 2022). In contrast, the exclusive dominance of *Georgenia* in $R1_{AQS-}$ reflects a population less competitive under conditions favoring mediated electron transfer and syntrophic specialization.

Overall, the taxonomic and functional patterns demonstrate that efficient and stable LAS removal was governed by synergistic interactions between microaeration and AQS rather than by microbial diversity or methane content alone. Microaeration expanded ecological niches and enhanced metabolic robustness, while AQS selectively facilitated electron exchange between LAS-degrading bacteria and methanogenic archaea. The convergence of these effects in $R2_{AQS+}$ promoted a microbial community highly specialized for coordinated electron flow, explaining the superior LAS removal, enhanced COD degradation, and increased methane production observed under this condition.

3.4. Impact of anthraquinone-2-sulfonate (AQS) supplementation and microaeration

The improved LAS removal observed in the AQS-amended reactors corroborates the role of quinone-based redox mediators in facilitating the transformation of recalcitrant sulfonated compounds through enhanced extracellular electron transfer. Nevertheless, the performance of $R2_{AQS+}$ indicates that reactor redox conditions modulated the effect of AQS. While AQS addition alone ($R1_{AQS+}$) increased LAS removal without major shifts in microbial diversity, its combination with microaeration led to a microbial community characterized by higher richness, altered evenness patterns, and the enrichment of taxa capable of functioning under low-oxygen conditions. This configuration supports a degradation scheme in which microaeration expands metabolic functionality and ecological niches, whereas AQS improves electron transfer efficiency within this diversified community. Importantly, the absence of process instability or inhibition suggests that oxygen was rapidly consumed in non-methanogenic pathways, preserving anaerobic functionality.

Although direct identification of LAS intermediates was beyond the scope of this study, the consistent association between enhanced LAS removal, distinct microbial structuring, and combined redox manipulation provides process-level evidence that microaeration and AQS interacted synergistically to improve LAS degradation.

From an operational perspective, however, microaeration demands strict control. Oxygen must be dosed at levels sufficient for enzyme-level activation but low enough to avoid its escape to the headspace, where it would compromise biogas combustion safety by altering the flammability range. Maintaining this balance is challenging because real-time dissolved oxygen monitoring in anaerobic systems remains unreliable, and commercial DO sensors often exhibit poor sensitivity at the microaerobic range required for mediators and surfactant transformation. Thus, control strategies based on the system's redox potential become particularly attractive. Because AQS actively participates in redox cycling, monitoring the reactor's redox potential provides a more mechanistic and responsive basis for adjusting aeration, avoiding oxygen accumulation, and optimizing mediator performance.

Despite these advantages, the continuous addition of soluble AQS increases operational costs and generates concerns for downstream treatment, especially if residual mediator exits with the effluent. For full-scale applications, this limitation has driven interest in immobilized mediator systems. Activated carbon, biochar, and engineered nanomaterials containing quinone-rich functional groups can retain the mediator within the reactor, provide long-term electron-shuttling capacity, and substantially reduce the demand for repeated dosing (Zhang et al., 2020). Such materials also align with practical anaerobic treatment constraints by simplifying operation and lowering chemical consumption while preserving the enhanced LAS degradation observed in

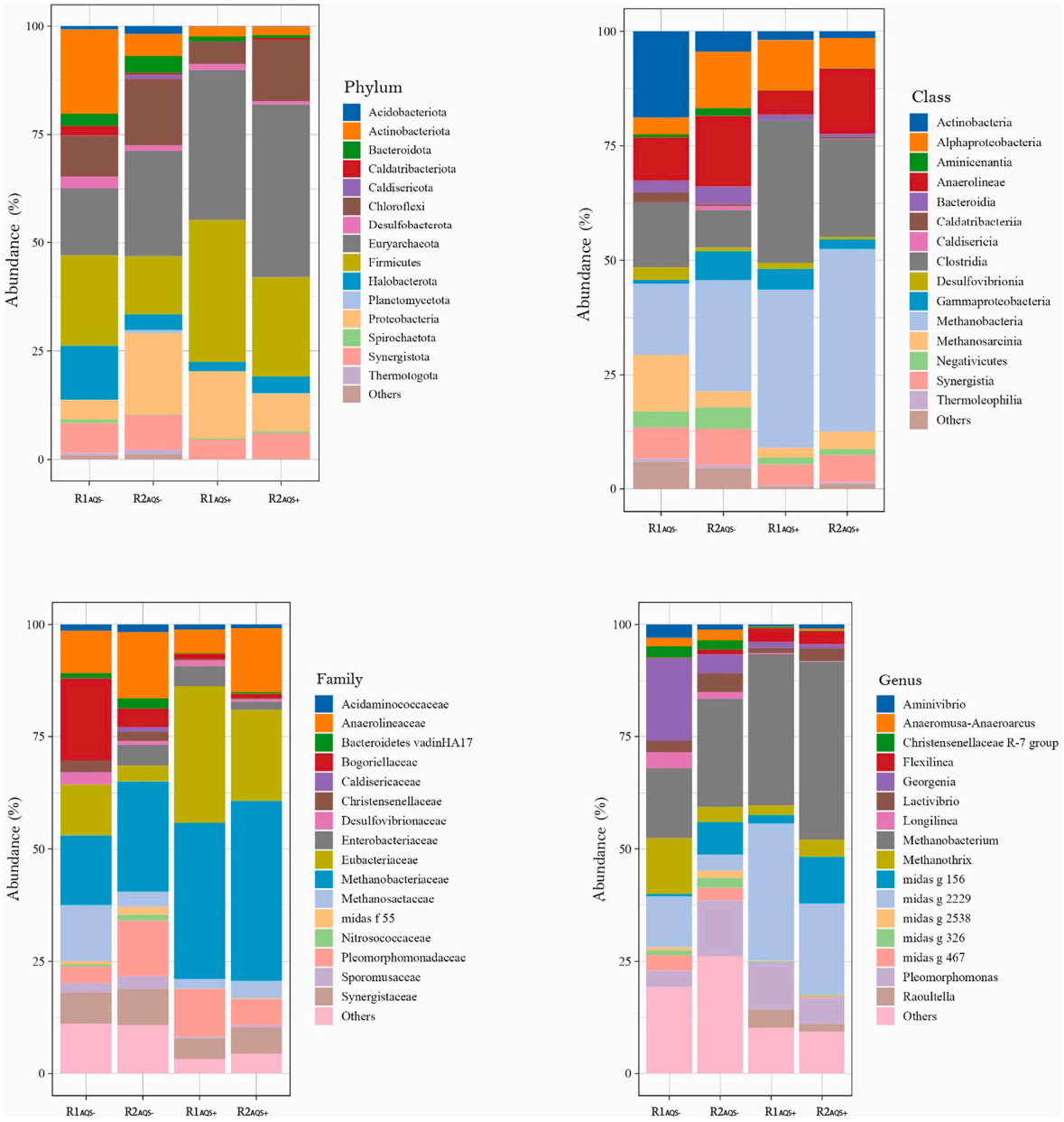


Fig. 8. Variation of bacterial community structure at the phylum, class, family, and genus level for UASB without AQS ($R1_{AQS-}$), microaerated UASB without AQS ($R2_{AQS-}$), UASB with AQS ($R1_{AQS+}$) and microaerated UASB with AQS ($R2_{AQS+}$).

the present study.

Overall, integrating AQS with microaeration offers a promising, operationally relevant approach to improving the removal of LAS and other personal-care-related pollutants while simultaneously enhancing biogas quality. The feasibility of this strategy, however, will depend on advances in microaeration control, redox-based monitoring, and cost-effective immobilization of redox mediators for long-term application.

4. Conclusions

This study demonstrated that the use of anthraquinone-2-sulfonate (AQS) as a redox mediator, both alone and in combination with microaeration, can enhance the performance of UASB reactors treating LAS-containing wastewater and treated at low HRTs. While the application of AQS under strictly anaerobic conditions did not significantly alter reactor stability or organic matter removal, it promoted the enrichment of methanogenic Archaea and selected syntrophic bacterial groups, suggesting the development of improved redox-driven biodegradation pathways.

The combination of AQS and microaeration ($R2_{AQS+}$) yielded the highest LAS removal, COD degradation, and methane production per unit of applied organic load, indicating a synergistic effect between redox mediation and controlled oxygenation. This condition also promoted a more diverse and archaeal-enriched microbial community, characterized by the dominance of *Methanobacterium* and *midas_g_2229*, potentially associated with enhanced LAS degradation under syntrophic conditions.

These findings highlight the potential of integrating redox-active compounds and low-intensity aeration as a viable strategy to improve anaerobic treatment of surfactant-rich wastewaters. Future studies should investigate the long-term stability of such systems, assess the fate of intermediate metabolites, and explore the functional roles of key microbial taxa identified, particularly through metagenomic or transcriptomic approaches.

CRediT authorship contribution statement

Maurício Guimarães de Oliveira: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vicente Elício Porfiro Sales Gonçalves da Silva:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis, Data curation, Conceptualization. **Steven Renato Ferreira Vasconcelos:** Investigation. **André Bezerra dos Santos:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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